

# Siqing Yuan

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## Education

- **ShanghaiTech University**, Shanghai, CN **09/2023 - 06/2026**  
Master of Engineering in Electronic and Information Engineering  
Supervisors: Prof. Dinggang Shen, Dr. Zhong Xue, Dr. Lei Chen  
**Cumulative GPA: 3.96/4.00**  
**Coursework:** Biomedical Imaging (A+), Medical Big-Data and Artificial Intelligence (A), Matrix computations (A), Neural Signal Processing and Data Analysis (A+), Medical Image Processing and Analysis (A), etc.
- **Northeastern University at Qinhuangdao**, Hebei, CN **09/2018 - 06/2022**  
Bachelor of Engineering in Measurement and Control Technology and Instrumental  
**Cumulative GPA: 3.21/4.00**

## Publication & Patent

- **Siqing Yuan**, Yulin Wang, Zirui Cao, Yueyan Wang, et al. *LGE-Guided Cross-Modality Contrastive Learning for Gadolinium-Free Cardiomyopathy Screening in Cine CMR*, Accepted at **MLMI 2025 (MICCAI workshop) (Oral Presentation)**.
- Zehao Weng, Dongdong Gu, Yuzhong Chen, **Siqing Yuan**, et al. *CoMoSeg: Anatomical Consistency and Cross Modality Guidance for Robust Brain Tumor Segmentation Using Partially Labeled MR Sequences*, Accepted at **CMMCA 2025 (MICCAI workshop)**.
- **Siqing Yuan**, Lei Chen, Zhong Xue, Dinggang Shen, *Contrastive Learning and Metadata-Guided Classification Network for Automatic Cardiomyopathy Diagnosis in Multi-Sequence CMR Images*, In preparation.
- **Siqing Yuan**, Lei Chen, Zhong Xue, Dinggang Shen, *Feature Space Augmentation for Class-Imbalanced Cardiomyopathy Diagnosis*, In preparation.
- Hui Wang, Lei Xu, **Siqing Yuan**, Lei Chen, et al. *Method, Device and Equipment for Cardiac Image Processing*. **Patent pending (Publication No. CN120339183A), as the lead student inventor.**
- Lei Chen, **Siqing Yuan**, Zhong Xue, Dinggang Shen. *Multimodal Medical Imaging Analysis Method and Computer Program Product*. **Patent pending (Application No. 2025107586905), as the lead student inventor.**
- **Siqing Yuan**, Lei Chen, Zhong Xue, Dinggang Shen. *Automated Method and System for Organizing Multimodal Cardiac MRI Data*. **Patent pending, as the lead student inventor.**

## Research & Project Experience

### Research on Intelligent Screening and Diagnosis System for Cardiomyopathy Based on Multi-sequence CMR Images (Functions as an independent algorithm researcher and technical support) **08/2024 - Present**

- Developed an automated pipeline to organize raw DICOM files into the required format the required format based on tag values, achieving a **fivefold increase in data organization efficiency**.
- Utilized nnU-Net to train a **cardiac segmentation model** that automatically extracts heart regions from CMR data, thereby improving the accuracy of subsequent classification tasks.
- Designed a **classification framework** utilizing multimodal fusion and contrastive learning, achieving a **5% improvement in accuracy** for cardiomyopathy diagnosis compared to current state-of-the-art models.
- Adapted the LayerCAM **visualization** technique to support multi-modal 3D inputs of the classification model; applied visualization tools, such as **t-SNE**, to evaluate performance and guide algorithm refinement.
- Design a **feature space enhancement method** to improve the generalization performance of the proposed classification model on datasets with varying distributions. (*In Progress*)

### Academic Seminar (co-organized by ShanghaiTech University and United Imaging Intelligence) **09/2023 - Present**

- Explored and compared **visualization techniques** for CNN-based network interpretability, such as CAM, Grad-CAM, Layer-CAM, and Score-CAM, focusing on their differences, strengths, and limitations.
- Analyzed and presented findings from an ECCV 2020 paper on **feature-space enhancement** for long-tailed data challenges and a MICCAI 2024 paper on enhancement for image cross-domain generalization.
- Engaged in peer-led discussions on **vision-language models** for medical report generation, **chain-of-thought** reasoning to improve report usability, and **agent-based pipelines** for orchestrating large and small models, etc.

### FastSurfer as an Alternative to FreeSurfer in HCP Minimal Preprocessing Pipeline **10/2023 - 12/2023**

- Replaced FreeSurfer with FastSurfer (a fast and accurate deep learning-based neuroimaging pipeline) in the HCP minimal preprocessing pipeline, significantly accelerating the processing workflow.
- Conducted qualitative analyses on surface reconstruction and cortical parcellation, along with quantitative analysis of runtime before and after the substitution, confirming the efficiency of the integration.

#### **Estimate A $\beta$ -PET Pattern from fMRI Data based on Deep Learning** **10/2023 - 12/2023**

- Developed a method utilizing **Graph Convolutional Network** to integrate low-order and high-order functional connectivity information obtained from rs-fMRI data to predict A $\beta$ -PET deposition patterns.
- Analyzed results from five-fold cross-validation and authored a conference-style paper exploring non-contrast imaging techniques as alternatives to contrast-agent sequences in predicting Alzheimer's disease biomarkers.

#### **Internship Experience**

##### **Shanghai United Imaging Intelligence Co., Ltd** **07/2024 – Present** **Algorithm Engineer, Department of Research and Development** **Shanghai**

- Assisted in clinical segmentation labeling at Ruijin Hospital; implemented an **active learning strategy** prioritizing the most uncertain and representative samples for manual correction, thus reducing redundant effort.
- Trained a pelvic segmentation model using **nnU-Net**; fine-tuned a **pre-trained VST network** to train a three-class classification model distinguishing among Alzheimer's patients, MCI subjects, and healthy controls.
- Assisted in integrating the cardiomyopathy diagnosis module into the existing software and **addressed various deployment challenges**, including code optimization for GPU memory limits in LayerCAM visualization.
- Develop a **product demo** for intelligent visualization of health reports. *(In Progress)*

#### **Competition Experience**

##### **ShanghaiTech University 7th Innovation and Entrepreneurship Conference** **07/2025**

*Project: AI-Based Multimodal Imaging System for Accurate Periodontal Health Assessment* **Second Prize**

- Designed an AI periodontal health assessment system utilizing existing patents and papers, capable of automatically detecting periodontal disease through panoramic X-rays, CBCT, and intraoral scanning images.

*Project: One-Stop Intelligent Post-Processing Analysis System for Cardiac MRI* **Third Prize**

- Designed a system for processing CMR images based on existing patents and research, using AI technology for automated segmentation, clinical metric calculation, and disease diagnosis in multi-sequence CMR imaging.

##### **Other Competition Achievements Unrelated to Medical Imaging** **2021 - 2022**

- 7th **Hebei Province** College Student Engineering Training Comprehensive Ability Competition **First Prize**
- 7th **Hebei Province** "Internet+" College Student Innovation and Entrepreneurship Competition **Bronze Award**
- 14th **National** College Student Energy Conservation and Emission Reduction Competition **Third Prize**

#### **Scholarship & Award**

- Second-class Academic Scholarship for Graduate Students (three times) **2023 - 2025**
- Outstanding Graduation Thesis at the University Level **06/2022**
- Innovation and Entrepreneurship Scholarship for College Students **06/2022**
- Second-class Scholarship (3%), Third-class Scholarship (20%) **2020 - 2021**
- Special Contribution Award from the Student Innovation and Entrepreneurship Association **10/2020**
- Excellent Student Leader, Outstanding Student Leader in Innovation and Entrepreneurship Work **2019 - 2020**

#### **Technical Skills**

- **Programming:** Python, LaTeX, MATLAB
- **Machine Learning:** PyTorch, nnU-Net, Video Swin Transformer, CAM, t-SNE, Scikit-learn
- **Data Handling:** pydicom, ITK-SNAP, Pandas, NumPy, Matplotlib

#### **CAMPUS & LEADERSHIP EXPERIENCE**

##### **Minister, Competition Department, University Innovation and Entrepreneurship Association** **2018 - 2020**

- Developed semester plans and coordinated departmental activities, including university-level selection for innovation competitions, crafting the student innovation handbook, and managing scholarship evaluations.

##### **Volunteer Activities** **2018 - 2020**

- Contributed as a volunteer for various initiatives, including the China College Students Programming Contest and the Hebei Marathon, totaling 100 hours of dedicated volunteer service.